

CMSC216: Course Mechanics

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Immediate Logistics

First course meeting is Lecture on Tue 01-Sep-2026

- ▶ Discussions **Do Not meet on Mon 31-Aug-2026**
- ▶ Discussion **will meet on Wednesday** to help students set up their environment and complete Lab 01

Overview of Mechanics: See Syllabus for Details

Lectures

- ▶ 2x per week, 75 minutes
- ▶ Chat, Exercises
- ▶ Earn Bonus Engagement Points
- ▶ All Lectures Recorded
- ▶ 3 Exams and a Final

Assignments: Gradescope

- ▶ Labs Weekly, 12 total, 10 for full credit
- ▶ Projects: 5 planned, larger programs
- ▶ Free Collaboration

ELMS/Canvas Homepage

Links to course schedule, staff contact, setup guide, surveys, etc.

Discussion

50min on Mon/Wed, get TA help on assignments, practice problems, some opportunities EPs

Engagement Points

- ▶ Bonus Credit for overall score
- ▶ Earn in Lecture, Discussion, Piazza, via on-time project submission

Office Hours

Mostly In-Person, Some Online via Zoom, see ELMS/Canvas for times + locations

Piazza Discussion board

Async Questions/Answers, Students ask, Staff Answer; Students answering other Students earns EPs

Lectures and Hot Seats

First Lecture is Tue 01-Sep

- ▶ Most lectures will take place in-person on campus
- ▶ **Some lectures** may take place on Zoom due to extenuating circumstances (illness, travel, family emergencies)
- ▶ During Lecture, Kauffman will have **exercises** for students to do during class
- ▶ Students will chat each other up about the exercises
- ▶ On resuming, chat about answers with 1-2 folks, usually volunteers and victims selected from the “Hot Seats” (first few rows of lecture hall or randomly when online)
- ▶ Showing effort earns **Engagement Points**
- ▶ Students are encouraged to ask questions when prompted
- ▶ **Lectures are recorded and posted** for students that can't make it to the synchronous meeting or want to review later

Communication

Piazza: Discussion Board

- ▶ Questions on course content / logistics / projects / etc.
- ▶ Announcements from Staff
- ▶ Read the Etiquette Post so you can post Answerable Questions

Gradescope

- ▶ Submit Labs and Projects
- ▶ Receive Exam Grades
- ▶ Request Regrades on submitted work

Email Kauffman for

- ▶ Appointments outside of office hours
- ▶ Personal issues such as illness, emergencies, accommodation requests, missing exams

Don't use ELMS/Canvas messages or Private Piazza posts - I won't always see them

You don't need to notify anyone if you just miss a lecture

Office Hours

- ▶ Office Hours are open to all students, no appointments needed
- ▶ Times and locations vary, listed on Canvas and usually solidify during the first week of classes
- ▶ Helps to come **prepared** to office hours: specific question, things you've tried to fix bugs; may lose time if not prepared

In-person Office Hours

- ▶ TAs are in IRB 1108; Kauffman in IRB 2226
- ▶ Usually have Queue: physical line or whiteboard / paper list

Online Office Hours

- ▶ Some staff may hold limited online office hours via Zoom
- ▶ Office hours schedule will include these along with a link to attend online office hours

Reading and Textbooks

Computer Systems: A Programmer's Perspective

- ▶ **3rd Edition** which covers 64-bit arch rather than 32-bit
- ▶ Author: R. Bryant and D. O'Hallaron,
- ▶ **REQUIRED:** it's expensive but an *excellent* text which will serve you well (if you read it)

C Programming

- ▶ Likely you'll want to do some reading on C programming to supplement in-class discussion
- ▶ *C Programming Language* Second Edition by Brian Kernighan and Dennis M. Ritchie,
 - ▶ **Optional:** not a bad read from the original authors of C
- ▶ **Free web resources** on C coding at the bottom of the Canvas front page
 - ▶ **Optional:** alternative to Kernighan and Ritchie

Lab01 and Programming Environment

First Discussion meeting is Wed 02-Sep

- ▶ TAs will introduce and cover some labs in Discussion
- ▶ Lab01 is designed to make sure that you are set up to program for the course
- ▶ Staff give demos/overview/tips then students work in groups or individually to solve exercises; Staff give help as needed
- ▶ Get Credit for lab by submitting completed work to Gradescope: can submit as a group
- ▶ Make sure to **ask for help during discussion** if you feel lost. We have *awesome TAs* who can get you over hurdles.

Lab01 Preparation

1. Look at the [Coding Environment Setup Guide](#) to learn how to access a Linux environment for coursework
2. Come in person to your assigned discussion time
3. Bring a laptop or other device capable of computing or plan to make a friend to work together

Prime Directive and Academic Integrity

PRIME DIRECTIVE: *Be able to explain your own work including homework code and exam solutions. The work you submit should be the product of your own effort and reflect your personal understanding.*

Assignment Regulations

- ▶ Projects and Labs are your chance to build your understanding.
- ▶ **Free Collaboration:** work with other students, AIs, browse the internet, construct solutions. **Make sure you can explain your work if asked;** practice explaining with peers and staff.
- ▶ Advice: Don't do assignments with AI. Practice doing things for yourself. *"If you're nothing without the suit, then you shouldn't have it."* -Tony Stark

Open Exam Regulations

- ▶ Exams are where you prove your **individual** understanding.
- ▶ CAN use notes, code you wrote, slides, textbook, practice materials, solutions provided by staff.
- ▶ CANNOT use AI, search engines, past exams from other students.

Appropriate Collaboration on Assignments

- ▶ Every problem presented in this class has been solved before by many others (OGs, your instructor, past students, etc.)
- ▶ Solutions exist online that can be located and those solutions can be regurgitated by your favorite LLM
- ▶ **The point of this and any class is to learn to do things yourself**, so you are **urged** to...
 1. Try solving problems yourself before prompting collaborators for their solutions; compare yours to theirs, explain both
 2. Get help from staff on appropriate ways to solve problems using functions and techniques discussed in class
 3. Practice explaining your answers to others (staff and students)
 4. Disable LLM/AI/Chat tools in your coding environment
 5. Own “your” answers: know the functions used, the algorithm implemented; if you can’t explain it, discard and rework
 6. Don’t give away code or ask someone to give away theirs. Collaborate, don’t copy, as you would in a team.

Exams will have problems you haven’t seen before. Train yourself to solve problems you’ve never seen before as that is what you’ll be paid to do or fired for being unable to do.

Policy: No Past Exam Sharing

- ▶ Students in this course may not obtain past exams from previous students
- ▶ Past students may not share their exams with present students
- ▶ All students in the course will be provided with practice exams to help them prepare for this semester's exams
- ▶ 6 students from Fall 2025 received XF grades for violating this policy; all claimed ignorance but a board of their peers found them responsible for their actions

Buy Why?

- ▶ Not everyone in the course has a friend from last semester who can share past resources so it is unfair for some students to get access while others don't
- ▶ Exams will be focused on the present semester's topics which may be different from the past which will be reflected in the practice exams provided

General Goals for the Course

- ▶ Include C and Assembly on your resume without reservations
- ▶ Understand how all high-level programming languages interact with the machine on which they run
- ▶ Outline what hardware is in a computing system and some basic principles that govern it
- ▶ Gain familiarity with the software abstractions of hardware that all Operating Systems provide
- ▶ Deepen debugging skills through working with a debugger and learning to fix problems in one's own code
- ▶ Become comfortable with working on the command line and on a remote computer

Expectations

Kauffman Can

- ▶ Provide guidance, entertainment, information, challenge
- ▶ Will do all of those in lecture, office hours, assignments, exams

Kauffman Cannot

- ▶ Force you to pay attention, do your HW, attend labs, read, ask when you don't know, practice, learn.
- ▶ Cannot force you to **CARE**, the critical factor in any endeavor.
- ▶ Caring leads to effort. Effort leads to improvement. Constant improvement leads to success.

Kauffman's Expectation

- ▶ You care at least a little bit and will cultivate an attitude of curiosity and engagement
- ▶ You will put some effort into our time together as I have

Do you Care?



This is my big gripe with LLMs in general: that LLMs don't care.. they can't care... they do what they're told, maybe, whereas like a junior [dev], they care. They go "oh, actually it was off by one pixel so I actually just went into the code base without anyone telling me and I aligned it better"... and you're like, "Hey, you care! I love it that you care!"*

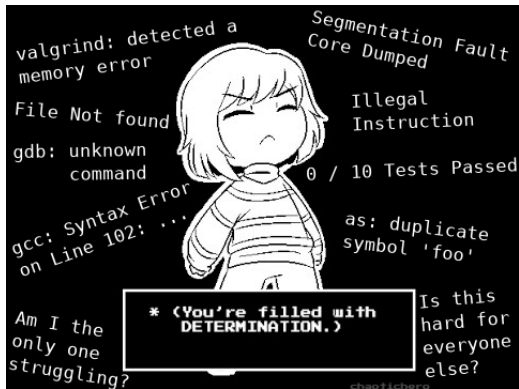
ThePrimeagen (Michael B. Paulson),
"Devin Is A Lie?"

Using a forklift to move weights will not make you stronger. Riding a horse will not make you run more quickly. Copying and pasting someone's story will not make you a better writer.

*Why would you pay for a computing course to train you and then let a tool rob you of the practice? **You won't if you care.***

*LLM: Large Language Model, an AI tool tuned to produce natural language and code answers to queries.

Don't Give Up, Stay Determined!



Students have different experience levels. Some have lots and make things look easy. For others, everything is new and intimidating. No one knows all of this stuff. Everyone struggles at some point. Get help from the staff. Support each other. Your peers will remember when you help them move forward and when you cut them down.

Respect and learn from one another.